Project Summary: Multigraded Homological Algebra & Geometry

Overview.
This proposal advances the study of multigraded homological algebra, algebraic geometry, and combinatorics. It has three integrated projects: i) establishing connections between minimal graded free resolutions and geometry on stacky curves and toric stacks, ii) clarifying the connection between multigraded Castelnuovo–Mumford regularity and free and virtual resolutions on toric varieties, and providing bounds and algorithms for multigraded regularity, and iii) building a uniform framework to study homological invariants across all toric varieties with the same Cox ring. The PI expects the outcomes of this proposal to include new theorems linking homological algebra and toric geometry, developing open-source software, and concrete actions to expand the mathematical community.

Intellectual Merit.
The first project seeks to generalize classical results about the syzygies of algebraic varieties, e.g., Green's N_p-theorem for curves of high degree, Green's conjecture concerning the syzygies of canonical curves, Ein and Lazarsfeld's work on asymptotic syzygies, to stacky curves and toric stacks. As part of this work, the PI proposes a number of concrete questions developing a suitable theory of kernel bundles and Koszul cohomology adapted to the multigraded setting of toric stacks and stacky curves, studying non-standard graded analogs of Koszulness and their implications for the multigraded Betti numbers of stacky curves. The PI plans to develop software for computing with toric stacks and expand the database of syzygy computations she maintains.

The PI's second project looks to further our understanding of multigraded Castelnuovo–Mumford regularity, and other measures of algebraic and geometric complexity, in the multigraded and toric settings. As part of this, the PI seeks to continue and generalize her recent work showing the precise relationship between free resolutions, virtual resolutions, and multigraded regularity on products of projective spaces to arbitrary toric varieties. Further, the PI suggests a number of directions seeking to develop meaningful bounds on multigraded regularity, and study how it behaves asymptotically. As a consequence of this work the PI hope to develop algorithms for effectively computing multigraded regularity.

The third project expands upon a developing theme that many homological properties of toric varieties only depend on the underlying graded ring. The main direction of this project is developing uniform homological algebra tools (multigraded regularity and virtual resolutions) in the D_{Cox} category. Using these tools, the PI hopes to prove: i) uniform vanishing and regularity statements that hold for all toric varieties with a fixed Cox ring, ii) descriptions of controlled changes in multigraded Betti numbers under variation of GIT stability, and iii) relate the local cohomology of modules after saturating by different monomial ideals corresponding to sheaves on different quotients. This project is closely connected to the first two projects.

Broader Impacts.
The PI will continue her work to broaden the mathematical community and develop the next generation of mathematicians. This includes continuing mentoring two undergraduates and three graduate students, organizing the *Unspoken Rules* seminar providing professional development to graduate students facing additional career challenges, and leading a project at a summer undergraduate research program in 2026. The PI served in a number of roles seeking to support the growth and promotion of the mathematical community. This has included serving on committees for the AMS and SLMath, and being the inaugural president of *Spectra*. The PI will continue serving on AMS committees, and plans to re-engage in organizing with *Spectra*.

The PI has organized several conferences: *AGNES* (2024), *GEMS of Commutative Algebra* (2019, 2023), Spec($\overline{\mathbb{Q}}$) (2022), *WAGS* (2021), *GEMS of Combinatorics* (2023), *Geometry & Arithmetic of Surfaces* (2019), and *M2@UW* (2018). The PI plans to organize *GEMS of Commutative Algebra* again in Fall 2026, and is preparing a proposal for a conference on syzygies and geometry at BIRS.